

Math 526, S06
Quiz 11

Name:
SSN: xxx-xx-

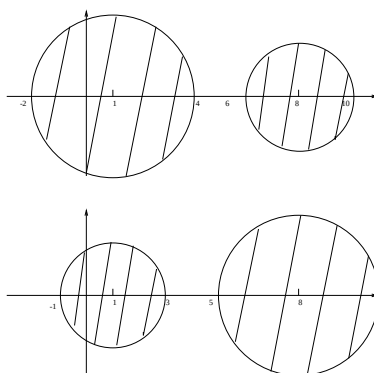
Instructions: There are totally 1 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (20 points) Let $\mathbf{A} = \begin{bmatrix} 8 & -2 \\ 3 & 1 \end{bmatrix}$,

a) Use the Gerschgorin Circle Theorem to show that $\mathbf{A}$ has a dominant eigenvalue and give a rough estimate for the location of this dominant eigenvalue.

Solution:

$$a_{11} = 8, \quad a_{22} = 1, \quad r_1 = 2, \quad r_2 = 3, \quad c_1 = 3, \quad c_2 = 2$$



From the above figure, it is easy to see that A has a dominant real eigenvalue λ_1 such that $6 \leq \lambda_1 \leq 10$.

b) With the starting vector $\mathbf{x} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, compute the first three loops of the power method for the matrix $\mathbf{A}$.

Solution:

1st loop:

$$\mathbf{z} = \frac{\mathbf{x}}{\|\mathbf{x}\|_2} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{x} = \mathbf{A}\mathbf{z} = \begin{bmatrix} 8 & -2 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 8 \\ 3 \end{bmatrix}, \quad \alpha = \mathbf{z}^t \mathbf{x} = 8.$$

2nd loop:

$$\mathbf{z} = \frac{\mathbf{x}}{\|\mathbf{x}\|_2} = \begin{bmatrix} 0.9363 \\ 0.3511 \end{bmatrix}, \quad \mathbf{x} = \mathbf{A}\mathbf{z} = \begin{bmatrix} 8 & -2 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 0.9363 \\ 0.3511 \end{bmatrix} = \begin{bmatrix} 6.7884 \\ 3.1601 \end{bmatrix},$$

$$\alpha = \mathbf{z}^t \mathbf{x} = 7.4658.$$

3rd loop:

$$\mathbf{z} = \frac{\mathbf{x}}{\|\mathbf{x}\|_2} = \begin{bmatrix} 0.9363 \\ 0.3511 \end{bmatrix}, \quad \mathbf{x} = \mathbf{A}\mathbf{z} = \begin{bmatrix} 8 & -2 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 0.9066 \\ 0.4220 \end{bmatrix} = \begin{bmatrix} 6.4086 \\ 3.1418 \end{bmatrix},$$

$$\alpha = \mathbf{z}^t \mathbf{x} = 7.1359.$$